A Neural Signature of Neuroticism?
Exploring the Functional Connectome to Predict Personality

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I. Introduction

Background / Rationale

Neuroticism:
A personality trait that serves as a **general vulnerability factor**, predicting a wide range of **mental disorders** (e.g., disorders of anxiety, mood, eating, substance use, and personality) as well as **physical health problems** (e.g., cardiovascular, respiratory, and gastrointestinal conditions) [1]

Functional Connectivity:
Studies investigating the **neural correlates** of neuroticism indicate that resting-state functional connectivity (RSFC) can **successfully predict** individual differences in the trait [2,3]. However, findings remain **inconsistent** [4], most studies rely on **static** connectivity approaches, and neuroticism is rarely modeled as a **latent construct.**

Aim and Relevance of This Study

- Integrate dynamic connectivity and latent trait modeling into neuroticism research
- Clarify the relationship between neuroticism and RSFC
- Identify neural signature of neuroticism to inform clinical prevention and treatment

Research Questions

Does **a latent variable model** adequately capture the structure of **neuroticism** in the Human Connectome Project (HCP) young adult dataset?

To what extent do global **static** and **dynamic** measures of RSFC **predict latent neuroticism traits** in healthy young adults?

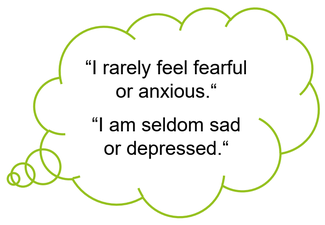
II. Methods

Measurement Model

Data and Instrument:
- **Sample**: *N* = 1198 healthy young adults (HCP-YA, 2025 data release) with a mean age of 28.84 years (range: 22-37)
- **Scale**: NEO-Five Factor Inventory (NEO-FFI) Neuroticism subscale (12 items; see thought bubble for two (reverse-keyed) example items).

Measurement Model Selection:
- **Estimation**: Utilized item response theory (IRT) via the *mirt* package to identify optimal latent structure
- **Comparison**: Evaluated four competing models
- **Selection Criteria**: χ^2 Likelihood Ratio Tests, AIC/BIC, and factor loading magnitudes

Model Approximation:
The optimal IRT structure was approximated via Confirmatory Factor Analysis (CFA) using the *lavaan* package



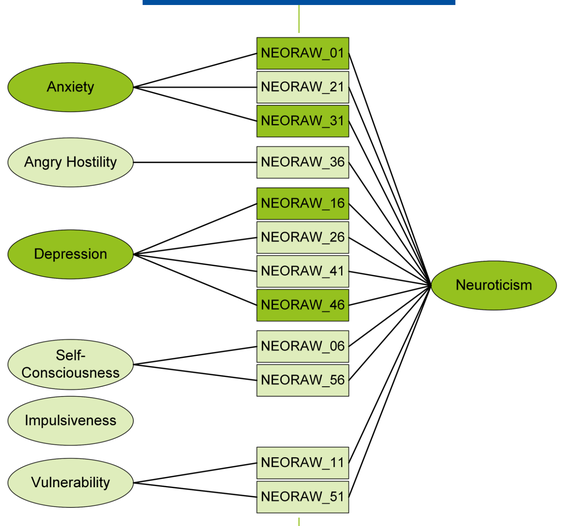


Figure 1: Allocation of items to specific neuroticism facets (left) and the general neuroticism factor (right). Non-transparent latent factors represent the components retained in the final bifactor model, while non-transparent items indicate those used to identify the specific facets. Notably, the general neuroticism factor was identified using all 12 items.

III. Results

Selected Model:
- **Bifactor Structure**: General neuroticism factor (g) and two specific facets – anxiety (s_anx) and depression (s_dep) – each comprising their two strongest items
- **Superior Fit**: Outperformed unidimensional and more complex models

Latent Structure Properties:
- **Factor loadings**: All standardized loadings > 0.3
- **Model Fit**: Statistics were excellent across both frameworks

approximates

fit	mirt bifactor	lavaan CFA
RMSEA	0.059	0.059
SRMSR	0.051	0.039
TLI	0.966	0.925
CFI	0.973	0.941

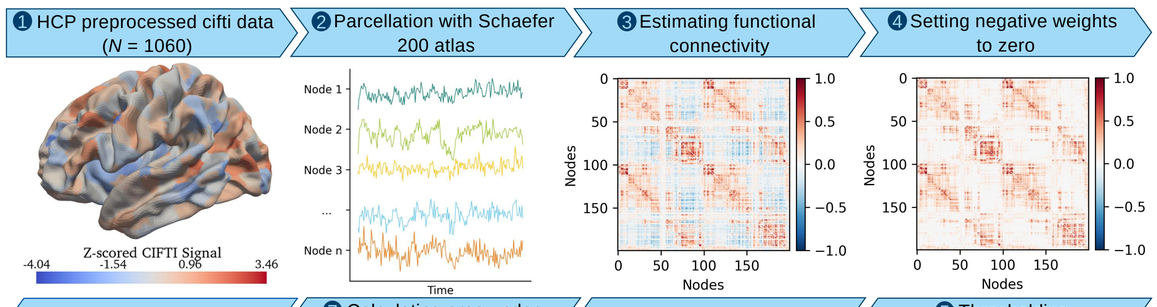
Table 1: Fit indices of chosen model

	factor loadings	mirt bifactor	lavaan CFA
g	NEORAW_01	0.459	0.390
	NEORAW_06	0.586	0.523
	NEORAW_11	0.705	0.637
	NEORAW_16	0.653	0.552
	NEORAW_21	0.702	0.612
	NEORAW_26	0.794	0.713
	NEORAW_31	0.639	0.541
	NEORAW_36	0.520	0.462
	NEORAW_41	0.752	0.670
	NEORAW_46	0.540	0.412
	NEORAW_51	0.671	0.588
	NEORAW_56	0.620	0.555
s_anx	NEORAW_01	0.465	0.394
	NEORAW_31	0.403	0.434
s_dep	NEORAW_16	0.394	0.432
	NEORAW_46	0.438	0.401

Table 2: Loadings of chosen mirt model and lavaan CFA

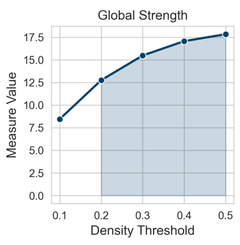
Brain Measures

1. HCP preprocessed cifti data (*N* = 1060)
2. Parcellation with Schaefer 200 atlas
3. Estimating functional connectivity
4. Setting negative weights to zero
5. Thresholding (density range 0.1 - 0.5)
6. Calculating global metrics
7. Calculating area under the curve (AUC)
8. Resulting brain measures



Static:
- Global Efficiency
- Small-World Propensity
- Global Clustering Coefficient
- Global Strength
- Betweenness Centrality
- Participation Coefficient

Dynamic:
- Mean BOLD Variability*,**
- Global Efficiency Variability
- Mean Edge Variability*



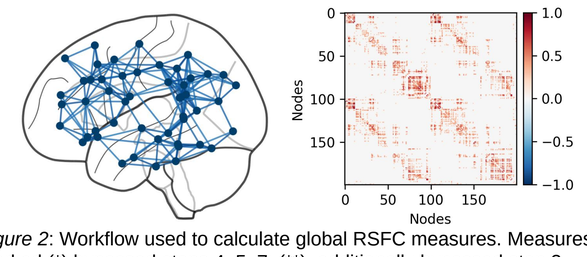


Figure 2: Workflow used to calculate global RSFC measures. Measures marked (*) bypassed steps 4, 5, 7; (**) additionally bypassed step 3.

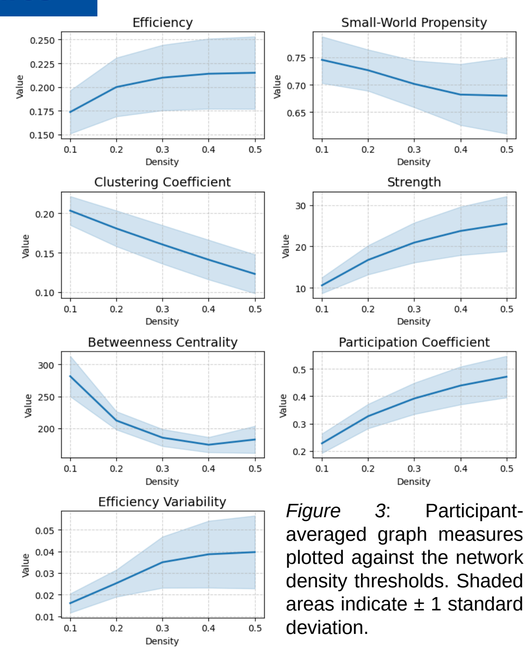


Figure 3: Participant-averaged graph measures plotted against the network density thresholds. Shaded areas indicate ± 1 standard deviation.

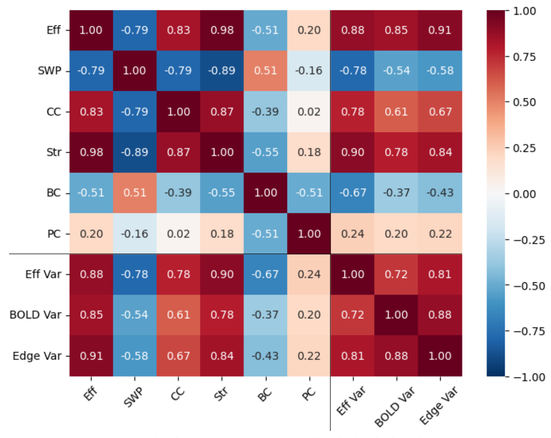


Figure 4: Correlation heatmap of global brain measures. Values represent the AUCs integrated across density thresholds 0.2–0.5. Static measures: Efficiency (Eff), Small-World Propensity (SWP), Clustering Coefficient (CC), Strength (Str), Betweenness Centrality (BC), Participation Coefficient (PC); Dynamic measures: Efficiency Variability (Eff Var), BOLD Variability (BOLD Var), Edge Variability (Edge Var).

Integrated SEM

Integrated Structural Equation Model (SEM):

- **Sample**: *N* = 1053 (intersection of participants providing both neuroticism data and brain imaging data)
- **Measurement Model Component:**
  Implemented the optimal bifactor structure identified in the preceding analysis (see "Measurement Model" section)
- **Structural Model Component:**
  - **Standardization**: All brain measures were z-transformed; for graph metrics, this utilized AUC values integrated across density thresholds (0.2-0.5)
  - **Predictor selection**: A subset of brain measures was selected based on the correlation matrix to reduce multicollinearity

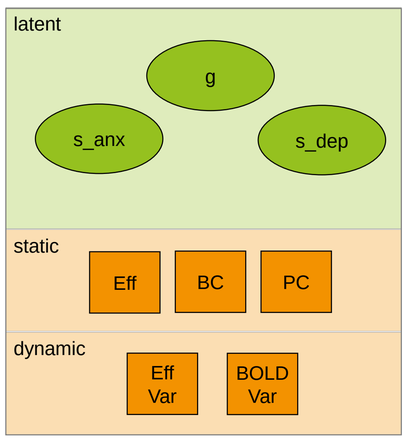


Figure 5: Illustration of included brain measures and latent variables

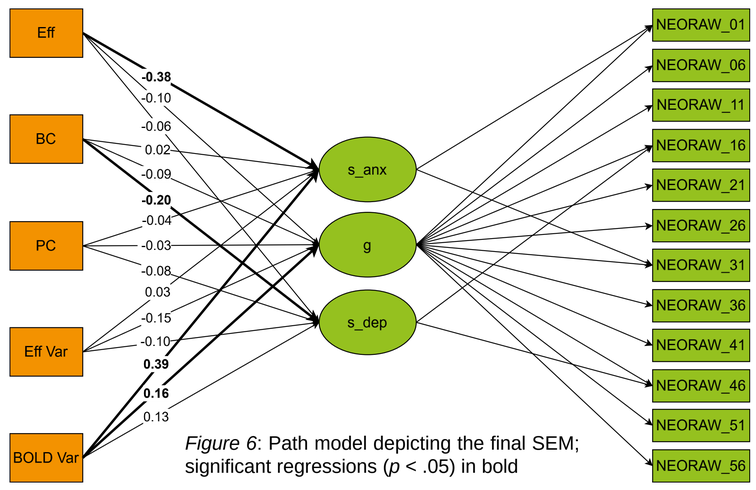


Figure 6: Path model depicting the final SEM; significant regressions (p < .05) in bold

fit	lavaan SEM
RMSEA	0.044
SRMSR	0.031
TLI	0.922
CFI	0.940

Table 3: Fit indices of the integrated SEM

latent factor	explained variance (R^2)
g	1.3 %
s_anx	4.5 %
s_dep	2.1 %

Table 4: Explained variance of the model's latent factors

IV. Discussion

Conclusion
- Excellent bifactor model fit supports a **multidimensional latent structure** for **neuroticism** in the HCP dataset
- Global RSFC markers (including **dynamic BOLD Var, static Eff**, and **BC**) significantly **associate with latent neuroticism**
- However, **collective impact** remains **negligible**, explaining 1-5% of total variance

Outlook / Limitations
- **NEO-PI-R** (48 items): Capturing all facets beyond the 12-item NEO-FFI subscale
- **Multiverse Analysis**: Testing robustness across varied atlases, connectivity methods, thresholds, etc.)
- **State-Based Metrics**: Incorporating e.g. switching rate to capture temporal fluctuations
- **Network-Level FC**: Shifting from global to system-specific measures to prevent masking divergent or opposite effects

Supplementary Materials
Scan QR code to access full documentation for:
- Neuroticism data preparation
- Tested measurement models
- Brain measure calculations
- SEM results

OPEN DATA (PA) OPEN MATERIALS

V. Acknowledgements

Data were provided by the Human Connectome Project, WU-Minn Consortium (Principal Investigators: David Van Essen and Kamil Ugurbil; 1U54MH091657) funded by the 16 NIH Institutes and Centers that support the NIH Blueprint for Neuroscience Research; and by the McDonnell Center for Systems Neuroscience at Washington University.
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VI. References

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